DSA

Practical No. 5 postfix evaluation

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Branch: CSE (AIML)

Batch: A2

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
#define MAX 50
int stack[MAX];
char post[MAX];
int top=-1;
void pushstack(int tmp);
void evaluate(char c);
int main()
{
  int i,l;
  //clrscr();
  printf("Insert a postfix notation :: ");
  scanf("%s",post);
  l=strlen(post);
  for(i=0;i<l;i++)
  {
    if(post[i]>='0' && post[i]<='9')
    {
       pushstack(i);
    }
    if(post[i]=='+' || post[i]=='-' || post[i]=='*' ||
```

```
    post[i]=='/' || post[i]=='^')
    {
       evaluate(post[i]);
    }
  }
  printf("\n\nResult :: %d",stack[top]);
  getch();
}

void pushstack(int tmp)
{
  top++;
  stack[top]=(int)(post[tmp]-48);
}

void evaluate(char c)
{
  int a,b,ans;
  a=stack[top];
  stack[top]='\0';
  top--;
  b=stack[top];
  stack[top]='\0';
  top--;
```

```
  switch(c)
  {
    case '+':
       ans=b+a;
       break;
    case '-':
       ans=b-a;
       break;
    case '*':
       ans=b*a;
       break;
    case '/':
       ans=b/a;
       break;
    case '^':
       ans=b^a;
       break;
    default:
       ans=0;
  }
  top++;
  stack[top]=ans;
}
```

/*

OUTPUT

Insert a postfix notation :: 95+7/3*4+

Result :: 10

*/